Differentially Expressed Genes Associated with Glaucoma

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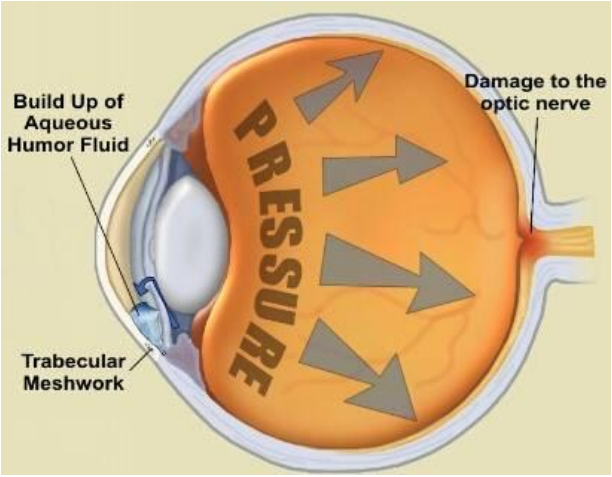
Introduction

What is glaucoma?

- Is a group of neurodegenerative diseases that affect the eye,
- Multifactorial. Aetiology not fully understood.
- Leading cause of irreversible visual loss globally. +3.6 million blind patients in 2020 (Steinmetz, 2021).
- 80 million people affected worldwide in 2020, with an estimation of 111 million by 2040 (Tham *et al*., 2014).
- Risk factors i.e., intraocular pressure (IOP), comorbidity, age, ethnicity.
- Silent disease.
- Short term therapeutic relief aimed to alleviate the IOP, the only modifiable risk factor.
- Novel diagnostic tools and therapeutic targets needed.
- Genetic implication gained significant attention in the last decade.

Normal eye

Eye with glaucoma



Adapted from Zoomax (2024).

Aims

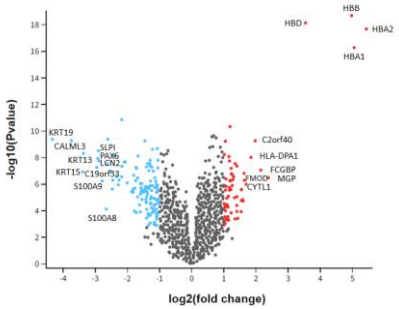
- Identify the most significant differentially expressed genes (DEGs) in glaucoma patients compared to a control group.
- Explore the involvement of these DEGs in the pathogenesis and progression of the disease.
- Identify new molecular targets with diagnostic and therapeutic potential.

Methods

GSE9944 – Astrocytes from optic nerve head
(9 glaucoma and 6 controls)

GEO[1] database

GSE27276 – Trabecular meshwork
(17 glaucoma and 19 controls)



Volcano plot

Analysis GEO2R

Limma R package
FDR[2]

Adjusted p-value ≤ 0.05
LogFC[3] ≥ 1, ≤ -1

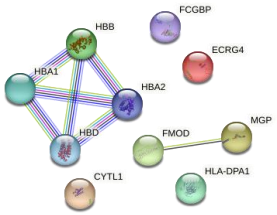
Top ten DEGs

Heat map

DEGs expression

STRING[4]

0.4 confidence
interaction score



PPI[5]

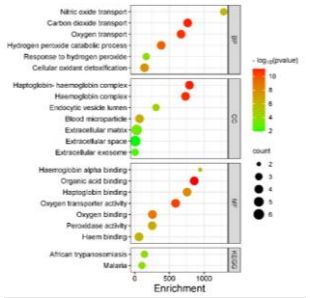
Protein interaction

DAVID[6]

Exact Fisher p-value ≤ 0.05
FDR

Enrichment analysis

Gene Ontology
KEGG[7]



[1]Gene Expression Omnibus
[2]False Discovery Rate
[3]Log Fold Change
[4]Search Tool the Retrieval of Interacting Genes/Proteins
[5]Protein-to-protein interaction
[6]Database for Annotation, Visualization and Integrated Discovery
[7]Kyoto Encyclopedia of Genes and Genomes.

Results

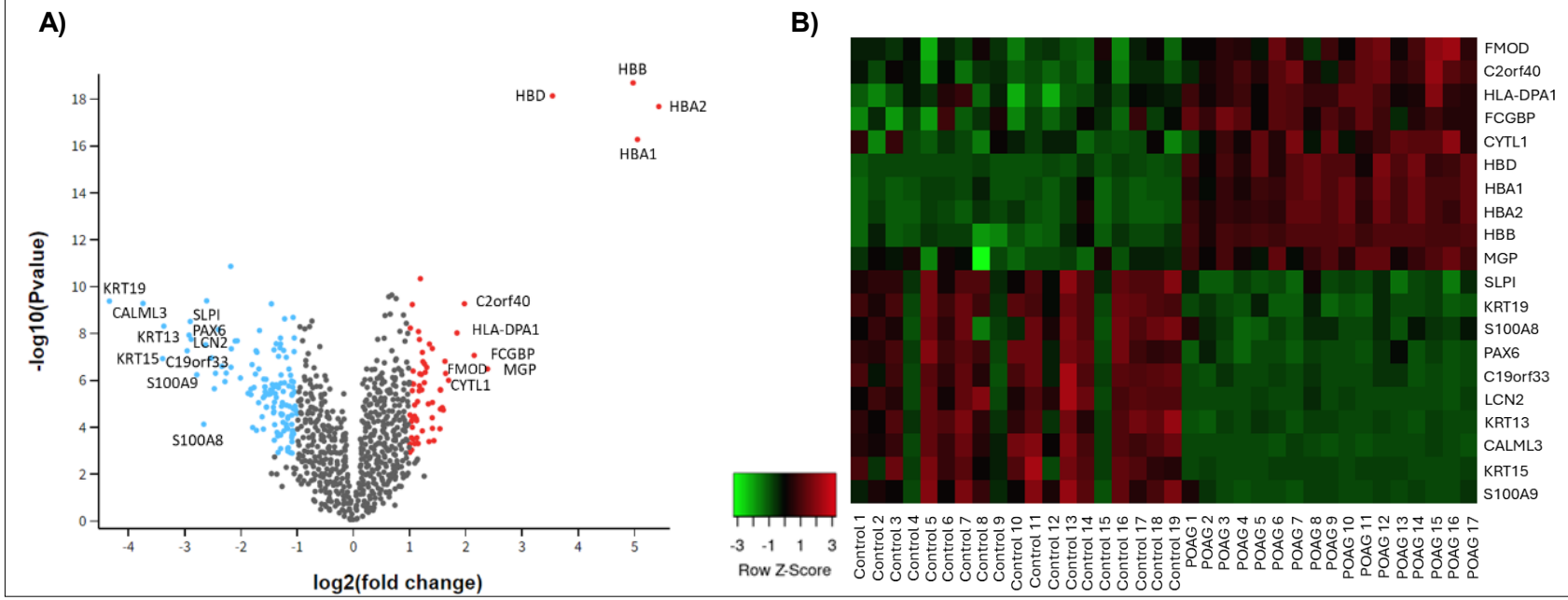


Figure 1. Trabecular meshwork (GSE27276).

A) Volcano plot. Upregulated DEG are in red and downregulated in blue. Black dots are unsignificant genes.

B) Heatmap. Expression level of the ten most significant upregulated DEGs in red and downregulated in green.

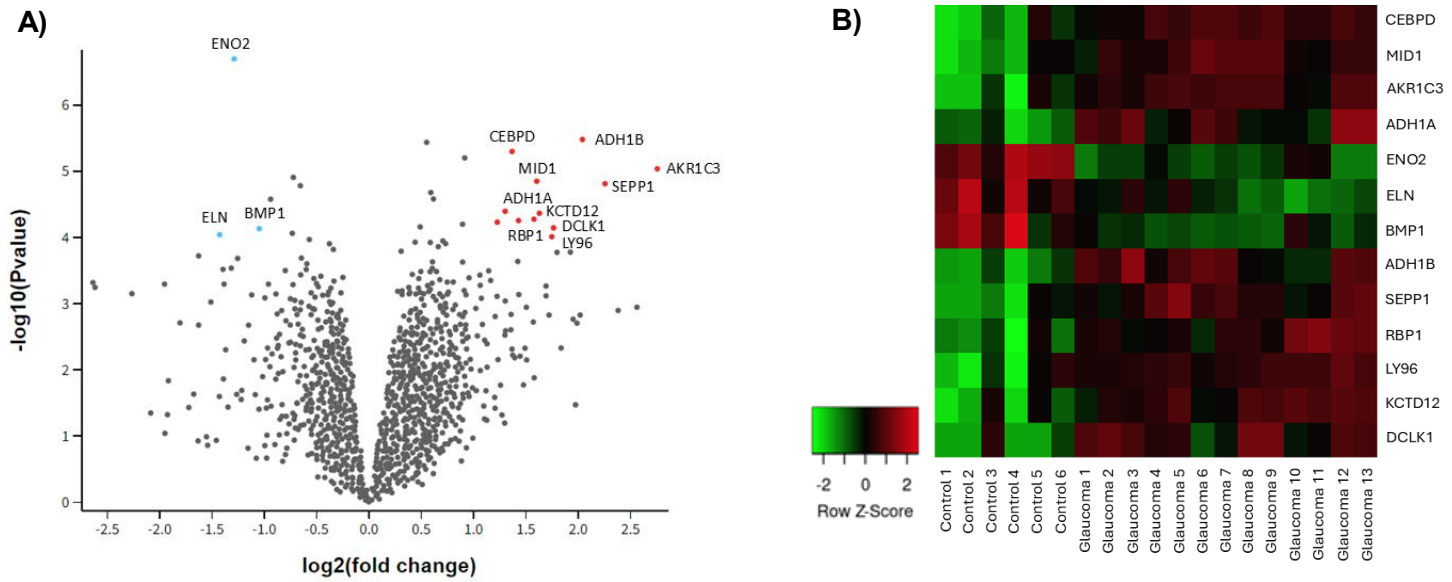


Figure 2. Optic nerve head astrocytes (GSE9944).

A) Volcano plot. Upregulated DEG are in red and downregulated in blue. Black dots are unsignificant genes.

B) Heatmap. Expression level of the ten most significant upregulated DEGs in red and downregulated in green.

Results

GSE27276 – Upregulated

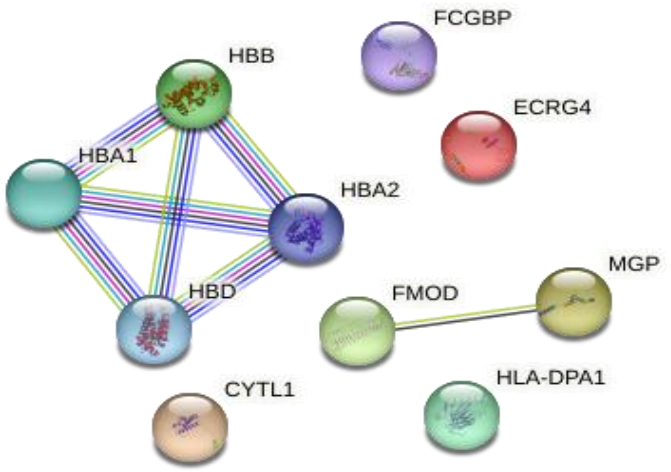


Figure 3. STRING PPI network of the top ten upregulated DEGs in the trabecular meshwork. Represented are two clusters. One composed of four nodes and six edges linking HBA1, HBB, HBD and HBA2. The other cluster is formed by FMOD and MGP linked by one edge. The remaining four DEGs, CYTL1, FCGBP, ECRG4 (C2orf40) and HLA-DPA1, were independent.

Each coloured circle (node) represents a DEG. Links (edges) between DEGs are the protein-to-protein interactions: known , predicted , and other type of interaction:

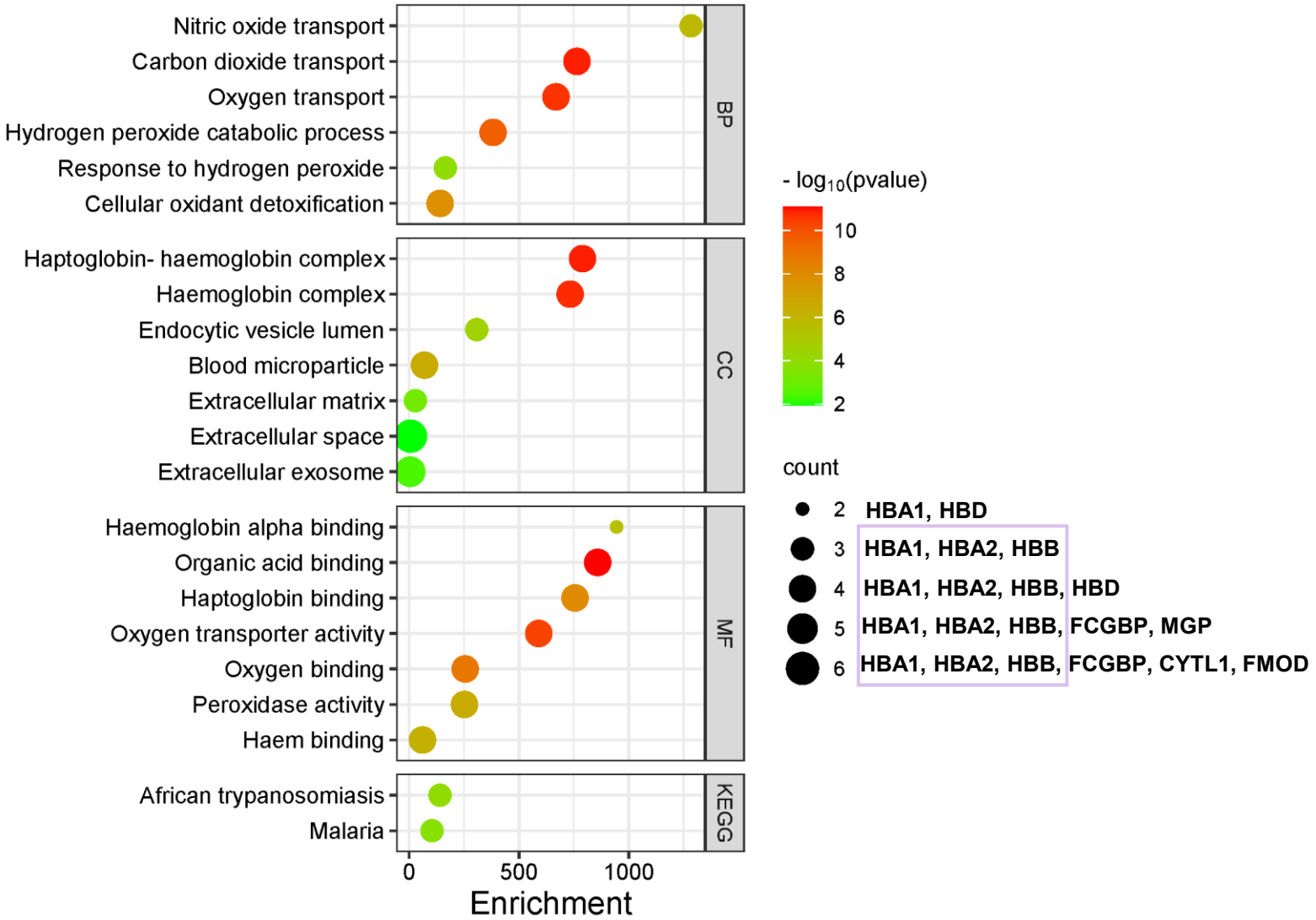


Figure 4. Functional enrichment analysis bubble plot of the ten most upregulated expressed genes in the trabecular meshwork. HBA1, HBA2 and HBB are the most represented genes in these terms. BP: biological processes, CC: cellular compartments, MF: molecular function, KEGG: Kyoto Encyclopedia of Genes and Genomes pathway. Count represents the number of genes involved in each process.

Results

GSE27276 – Downregulated

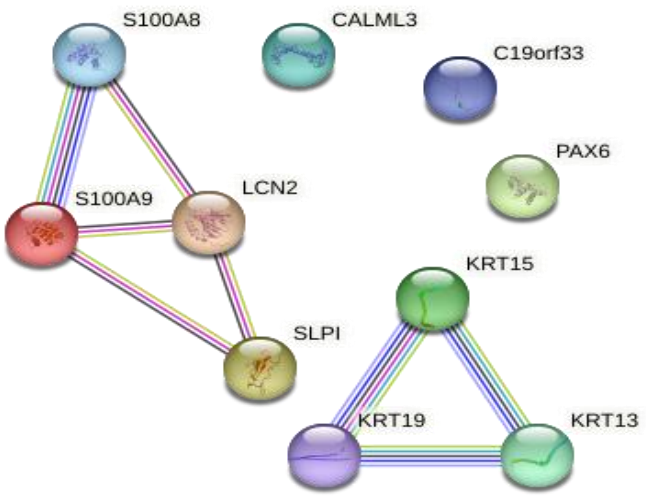


Figure 5. STRING PPI network of the top ten downregulated DEGs in the trabecular meshwork. Represented are two clusters. One with four nodes and five edges between S100A8, LCN2, SLPI and S100A9, being the latter also directly linked to LCN2. The other cluster consists of three nodes and three edges linking KRT15, KRT19 and KRT13. The remaining three DEGs, CALML3, C19orf33 and PAX6, were independent.

Each coloured circle (node) represents a DEG. Links (edges) between DEGs are the protein-to-protein interactions: known, predicted, and other type of interaction

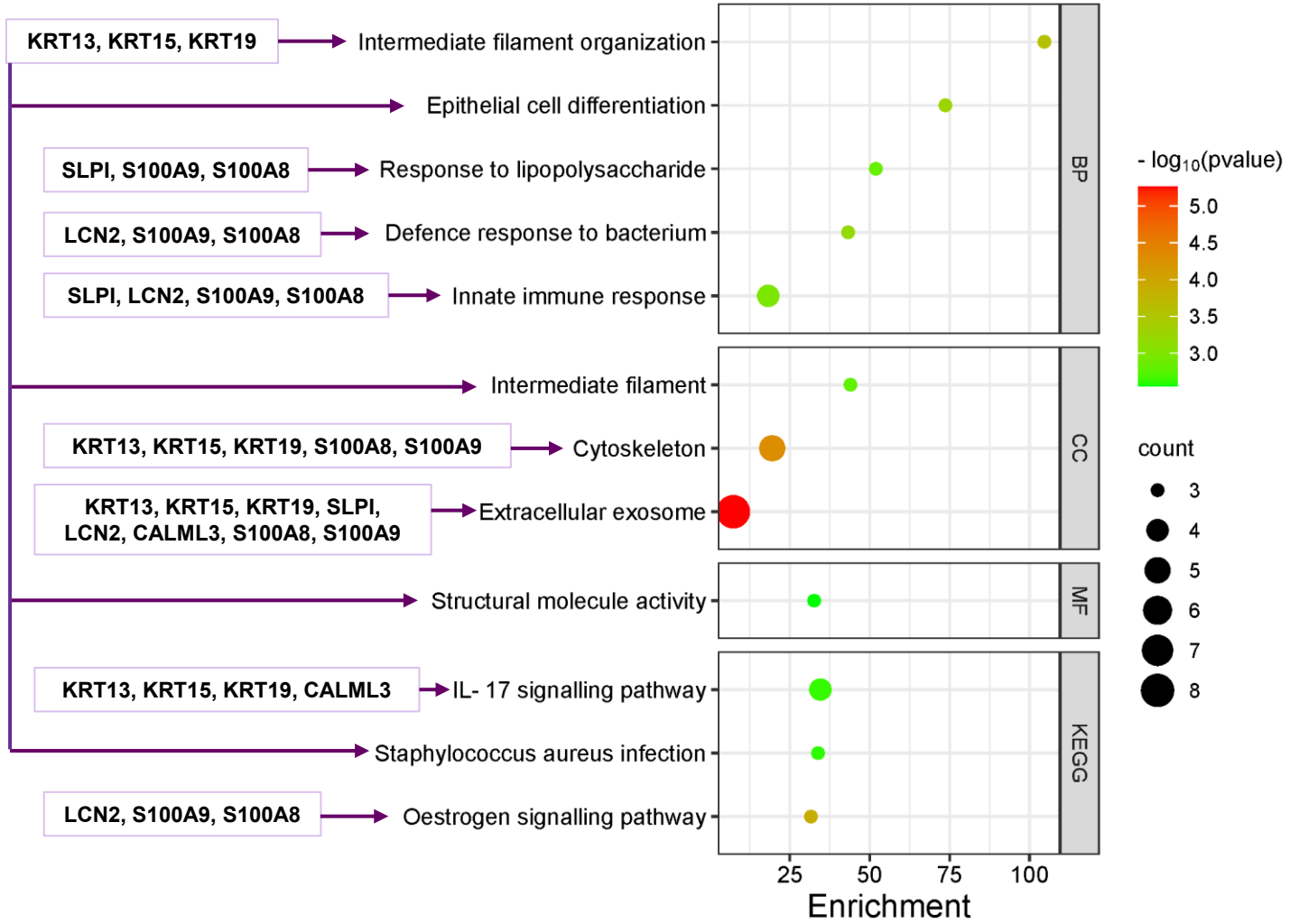


Figure 6. Functional enrichment analysis bubble plot of the ten most downregulated expressed genes in the trabecular meshwork. KRT13, KRT15, KRT19, S100A8 and S100A9 are the most represented DEGs in these terms. BP: biological processes, CC: cellular compartments, MF: molecular function, KEGG: Kyoto Encyclopedia of Genes and Genomes pathway. Count represents the number of genes involved in each process.

Results

GSE9944 – Upregulated

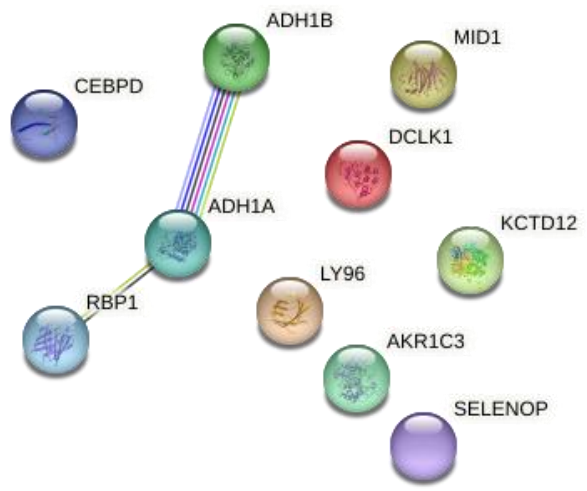


Figure 7. STRING PPI network of the top ten upregulated DEGs in the optic nerve head astrocytes. There is a cluster formed by three nodes and two edges; ADH1B and RBP1 linked to ADH1A. Seven nodes, CEBPD, MID1. CEBPD, MID1, DCLK1, LY96, AKR1C3, KCTD12 and SELENOP (SEPP1), were independent**.**

Each coloured circle (node) represents a DEG. Links (edges) between DEGs are the protein-to-protein interactions: known, predicted,and other type of interaction

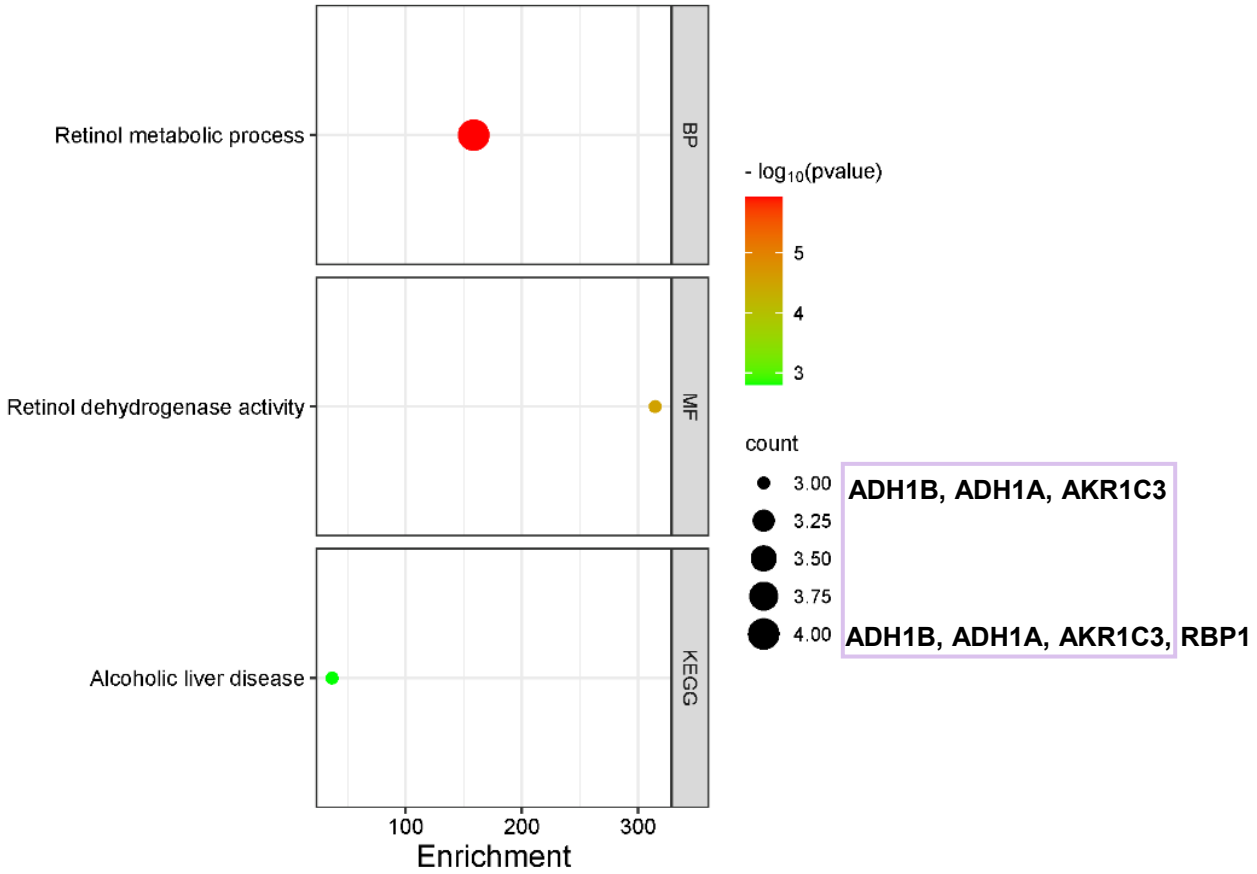


Figure 8. Functional enrichment analysis bubble plot of the ten most upregulated expressed genes in the astrocytes of the optic nerve head. ADH1B, ADH1A, AKR1C3 are the most represented DEGs in these terms. BP: biological processes, CC: celular compartments, MF: molecular function, KEGG: Kyoto Encyclopedia of Genes and Genomes pathway. Count represents the number of genes involved in each process.

GSE9944 – Downregulated

No significant functional enrichment detected.

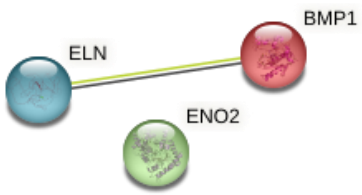


Figure 9. STRING PPI network of the downregulated DEGs in the optic nerve head astrocytes. Represented there is a cluster formed by two of the three significant downregulated DEGs, ELN and BMP1, while ENO2 remained independent. Each coloured circle (node) represents a DEG. Links (edges) between DEGs are the protein-to-protein interactions: known, predicted ,and other type of interaction

Summary

- Upregulated DEGs in the trabecular meshwork were enriched in processes related to **haemoglobin** (HBA1, HBA2, HBB, and HBD) and in **extracellular matrix processes** (FMOD, MGP).
- Downregulated DEGs in the trabecular meshwork were enriched in both **structural processes** within epithelial cells (KRT19, KRT15, and KRT13), and in **immunologic responses** (SLPI, LCN2, S100A8 and S100A9).
- Upregulated DEGs in the astrocytes of the optic nerve head were enriched in **metabolic processes** related to retinol (ADH1A, ADH1B, and AKR1C3).
- The regulation of these DEGs may have therapeutic applications. Prioritise CALML3, AKR1C3 and ENO2.
- Identified DEGs may serve as potential diagnostic markers (ENO2) (Jiang *et al*., 2021).
- More research needed.

Limitations

- Limited number of individuals per dataset.
- Lack of demographic data available i.e., drugs.
- Lack of experimental procedures.

Conclusion

- Significant differentially expressed genes were identified in glaucoma during this study.
- Potential therapeutic and diagnostic application.
- More research is needed.

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